

2. Dataset Inspection Report

Telco Customer Churn Analysis

1. Dataset Overview

The dataset contains customer information from a telecommunications company. It includes demographic information, services subscribed to, contract details, payment methods, monthly charges, total charges, and customer churn status.

The purpose of the dataset inspection was to understand the structure of the data, identify data-quality issues, and prepare the dataset for exploratory data analysis (EDA).

2. Dataset Size

The original dataset contained:

- **Rows:** 7,043
- **Columns:** 21

During the data-cleaning process, **11 records** containing blank values in the TotalCharges column were removed.

The final cleaned dataset contains:

- **Rows:** 7,032
- **Columns:** 21

3. Dataset Variables

The dataset contains the following variables:

Variable	Description
customerID	Unique customer identifier
gender	Customer gender

Variable	Description
SeniorCitizen	Indicates whether the customer is a senior citizen
Partner	Whether the customer has a partner
Dependents	Whether the customer has dependents
tenure	Number of months the customer has stayed with the company
PhoneService	Whether the customer has phone service
MultipleLines	Whether the customer has multiple phone lines
InternetService	Type of internet service
OnlineSecurity	Whether online security is subscribed to
OnlineBackup	Whether online backup is subscribed to
DeviceProtection	Whether device protection is subscribed to
TechSupport	Whether technical support is subscribed to
StreamingTV	Whether the customer has streaming TV
StreamingMovies	Whether the customer has streaming movies
Contract	Customer contract type
PaperlessBilling	Whether the customer uses paperless billing
PaymentMethod	Customer payment method
MonthlyCharges	Customer's monthly charges
TotalCharges	Total amount charged to the customer
Churn	Whether the customer left the company

4. Data Types Inspection

Most categorical variables were stored as object data types.

The main numerical variables included:

- SeniorCitizen
- tenure
- MonthlyCharges
- TotalCharges

An issue was identified with the TotalCharges column. Although it represents numerical values, it was initially stored as an object data type because some records contained blank values.

The column was converted to a numerical data type during the cleaning process.

5. Missing Values

The main data-quality issue was found in the TotalCharges column.

There were **11 records with blank/invalid TotalCharges values**.

These records were removed because a valid numerical value was required for the analysis.

After cleaning:

- TotalCharges was converted to numeric.
- 11 incomplete records were removed.
- The final dataset contained 7,032 records.

6. Duplicate Records

Duplicate records were checked during the inspection process.

Result:

No duplicate customer records were identified in the cleaned dataset.

7. Data Cleaning Actions

The following steps were performed:

1. Inspected the dataset dimensions and column names.
2. Checked the data types of all variables.
3. Checked for missing values.
4. Identified blank values in TotalCharges.
5. Converted TotalCharges from object to a numerical data type.

6. Removed 11 records containing missing/invalid TotalCharges.
7. Checked for duplicate records.
8. Confirmed that the final dataset contained 7,032 rows and 21 columns.
9. Verified that the cleaned dataset was ready for exploratory data analysis.

8. Data Quality Summary

Data Quality Check	Result
Original records	7,043
Final records	7,032
Variables	21
Records removed	11
Main data-quality issue	TotalCharges
Duplicate records	None identified
TotalCharges after cleaning	Numeric
Ready for EDA	Yes

9. Conclusion

The dataset was successfully inspected and prepared for exploratory data analysis.

The main data-quality issue was the presence of 11 blank/invalid values in the TotalCharges column. These records were removed, and the column was converted to an appropriate numerical data type.

The final cleaned dataset contains **7,032 customer records and 21 variables**, with no duplicate records identified. The dataset is therefore suitable for analyzing customer churn patterns and answering the project's business questions.